



KIRAN KRISHNA TUNUGUNTLA

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Address: Munich, Germany

[GitHub](#) [LinkedIn](#) [Portfolio](#)

WORK EXPERIENCE

Software Engineer

Jan. 2026 - Present

Setrade GmbH

Munich, Germany

- Built and operated forecast-integration and scheduling services on Ionos for the cloud controller platform powering 25+ battery energy storage sites, using Docker K8s
- Improved Observability and reduced the resolution time of high priority schedule failures to 4 hours.
- Key Technologies: PHP laravel framework, React Nodejs, Docker Containers, K8s, Ionos

Software Engineer (Minijob)

Dec. 2023 - Present

Ceyeborg GmbH

Munich, Germany

- Designed a GStreamer pipeline for direct frame capture from Balluff camera onto SIMA MLSoC, performing RGB→NV12 conversion on-device prior to inference for real-time pipe-crack detection; eliminated host-PC bottleneck in the capture-convert-forward chain, improving end-to-end throughput from 2 FPS to 5 FPS (2.5× gain).
- Successfully ported *flux.1/schnell* model onto iPhone 13 pro using optimization strategies like Palettization, Pruning and Quantization bringing model size down by 4X while maintaining accuracy of 97%
- Key Technologies: Coreml, Python, OpenCV, Pytorch, Gstreamer, Docker Containers

IoT Software Engineer

Nov. 2022 - Dec. 2025

Charge-V GmbH

Munich, Germany

- Responsible for charging station services that include Energy Management, Monitoring and Diagnostics, Charging Controller and User Interface.
- Improved the availability of the charging station to 98% by employing error handling mechanism and network stability.
- Successfully developed boosting feature by combining more than two charging stations to deliver power upto 240kW.
- Key Technologies: Python, C++, Nextjs, Django/FastApi, Pytest, Ionos, Redis, MQTT, OCPP, Modbus and InfluxDB/Grafana

Software Developer

Sept. 2020 – Oct. 2022

Polytech Wind Power Technology Germany GmbH

Munich, Germany

- Responsible for software infrastructure development, device provisioning and management, application deployment and monitoring.
- Integrated AWS IoT core for package registry, device provisioning and management
- Key Technologies: Python, pip/conda, Docker containers, AWS, MQTT, PostgreSQL and Grafana.

Software Developer

March 2020 – Aug. 2020

Fos4x GmbH

Munich, Germany

- Played a key role in the design, development, testing and deployment of system health monitoring pipeline on data processing units (DPUs).
- Significantly reduced the time spent on troubleshooting and helped isolate the issues with much more confidence on remote turbines.
- Key Technologies: C++, Python, pip/conda, Docker containers, AWS, MQTT, PostgreSQL and Grafana.

Working Student & Master Thesis

March 2018 – Feb. 2020

Fraunhofer EMFT

Munich, Germany

- **Thesis title:** "Detection and Classification of Semiconductor Processes using Deep Learning Neural Networks"
- Achieved a 15% performance boost by hybrid models over non-hybrid models. Also, achieved a 13.5% overall performance boost by non-hybrid and hybrid models using data augmentation.
- Developed RNN architectures for energy consumption forecasting; data pipeline built with NumPy, Pandas and InfluxDB.

EDUCATION

Master of Science <i>Communications Engineering</i> Technical University of Munich Grade: 2.1	Oct. 2016 – Aug. 2019 Munich, Germany
Bachelor of Technology <i>Electronics and Communication Engineering</i> Vellore Institute of Technology Grade: 1.54	Aug. 2012 – May 2016 Vellore, India

HONORS, AWARDS AND CERTIFICATIONS

Autonomous Mobile Robots Received a <u>certificate</u> of completion from ETH, edx.org	Oct. 2019 - Jan. 2020
Awarded third position at Munich Tech Fest 2019 Sandkraft a unique solution to tackle air pollution in cities.	Nov. 2019
Deep Learning Specialization Received a <u>certificate</u> of completion from deeplearning.ai, Coursera	Mar. 2019 - June 2019
Infineon Summer School, Digital industry and Products Received a certificate of completion with distinction	Aug. 2018
Best technical workshop in gravitas'14 Awarded as one among the best technical workshops during gravitas'14 technical fest at VIT University	Sept. 2014

SOFTWARE SKILLS

Programming Languages:	Python(Library development), C/C++, Typescript, Shell/lua scripts.
Frameworks:	Django/FastAPI, React/Nextjs, Orocos-rtt.
Agentic coding tools:	Claude code, Github copilot.
Network Protocols:	TCP/IP, HTTP/HTTPS.
Technologies:	MQTT, CAN, Modbus, OCPP, Websockets.
Version Control:	GIT.
DevOps:	GitLab workflows, GitHub actions, Ansible, Podman/Docker, Kubernetes.
Cloud:	AWS, Ionos
Databases:	Redis, MySQL, PostgreSQL, NoSQL.

LANGUAGE SKILLS

English:	Fluent
Telugu:	Native
Hindi:	Intermediate
German:	Intermediate